

ProjectHub - 3 Module API, JWT Workflow, and Spring Boot Annotations

Base URL: <http://localhost:8081> | Modules: Student, Faculty, Admin | Includes auth/button-click APIs

Spring Boot Annotations Used

Annotation	Use in this project
@SpringBootApplication	Starts the Spring Boot application.
@RestController	Controller classes return JSON/text API responses.
@RequestMapping	Base paths: /users, /projects, /certificates, /admin.
@GetMapping / @PostMapping / @PutMapping / @DeleteMapping	HTTP endpoint mappings.
@RequestBody	Reads JSON request body.
@PathVariable	Reads path ids like {id}, {studentId}, {fileName}.
@RequestParam	Reads query params and multipart file fields.
@CrossOrigin("**")	Allows React frontend API calls.
@Service	Business logic classes: UserService, ProjectService.
@Autowired	Injects services, repositories, JWT utility/filter.
@Configuration / @Bean	Security, password encoder, OpenAPI beans.
@Component	JWTUtil and JWTFilter Spring components.
@Entity / @Table / @Id / @GeneratedValue	MySQL JPA entities.
@Document / @Id	MongoDB documents.
@OneToOne / @ManyToOne / @JoinColumn	Entity relationships.
@Valid / @NotBlank / @NotNull / @Size	Validation for project requests/models.
@Transactional	Atomic update/delete operations.
@RestControllerAdvice / @ExceptionHandler	Global API error handling.
@Query / @Param	Custom repository query.
@Value	Reads frontend URL and mail config.

JWT Workflow

- 1 POST /users/login receives email and encrypted password.
- 2 UserService normalizes email, decrypts password if prefixed with enc:v2, and verifies BCrypt hash.
- 3 JWTUtil creates HS256 token with subject=email, role claim, issued time, and one-hour expiry.
- 4 React stores token, userId, role, and fullName in localStorage.
- 5 Protected fetch calls send Authorization: Bearer <token>.
- 6 JWTFilter skips public APIs: login, forgot/reset password, register, /landing/click, OPTIONS, Swagger, file download.
- 7 JWTFilter validates token, extracts role, creates ROLE_<role>, and sets SecurityContext.

8 SecurityConfig enforces STUDENT, FACULTY, ADMIN, authenticated, and public rules.

Common Auth and Button-Click APIs

Action	Method	API	Auth	Purpose
Sign In / Enter	POST	/users/login	Public	Login and return JWT.
Forgot password click	GET	/landing/click?action=forgot_password_click	Public	Button-click tracking.
Back to Sign In	GET	/landing/click?action=back_to_signin_click	Public	Button-click tracking.
Send Reset Link	POST	/users/forgot-password	Public	Create reset token/send link.
Open Reset Link	GET	/landing/click?action=reset_password_link_click	Public	Button-click tracking.
Change Password / Enter	POST	/users/reset-password	Public	Reset password with token.
Reset Back to Sign In	GET	/landing/click?action=reset_back_to_signin_click	Public	Button-click tracking.
Register	POST	/auth/register	Public	Create user by role.

Module 1 - Student APIs

Button/screen action	Method	API	Role/Auth	Purpose
Dashboard	GET	/projects/analytics/student/{userId}	STUDENT/ADMIN	Analytics cards.
Dashboard	GET	/projects/mongo/student/{userId}	Authenticated	Project list.
Dashboard	GET	/certificates/student/{userId}	STUDENT/ADMIN	Certificate list.
Profile load	GET	/users/student/profile/{userId}	STUDENT/ADMIN	Student details.
Submit Project page	GET	/users/filter/role?role=FACULTY	Logged in	Faculty dropdown.
Submit Project page	GET	/deadline-rules/type/Project	Logged in	Deadline rules.
Submit Project file	POST	/projects/upload	STUDENT/ADMIN	Upload file.
Submit Project button	POST	/projects/mongo	STUDENT/ADMIN	Create project.
Edit Project save	PUT	/projects/mongo/update/{id}	STUDENT/ADMIN	Update project.
Delete Project	DELETE	/projects/mongo/{id}	STUDENT/ADMIN	Delete project.
View/download project	GET	/projects/download/{fileName}	Public	Open file.
Certificate page	GET	/deadline-rules/type/Certificate	Logged in	Certificate deadlines.
Certificate file	POST	/projects/upload	STUDENT/ADMIN	Upload certificate file.
Submit Certificate	POST	/certificates	STUDENT/ADMIN	Create certificate.
Edit Certificate	PUT	/certificates/{id}	STUDENT/ADMIN	Update certificate.
Delete Certificate	DELETE	/certificates/{id}	STUDENT/ADMIN	Delete certificate.
Profile Save	PUT	/users/student/profile/{userId}	STUDENT/ADMIN	Update profile.
Create notification	POST	/users/notifications	Logged in	Create notification.
Notifications	GET	/users/notifications/{userId}	Logged in	Load notifications.
Mark read	PUT	/users/notifications/read/{id}	Logged in	Mark notification read.
Delete notification	DELETE	/users/notifications/{id}	Logged in	Delete notification.

Module 2 - Faculty APIs

Button/screen action	Method	API	Role/Auth	Purpose
Dashboard	GET	/projects/mongo/faculty/{userId}	FACULTY/ADMIN	Assigned projects.
Dashboard	GET	/certificates/faculty/{userId}	FACULTY/ADMIN	Assigned certificates.
Assigned Students	GET	/users/faculty/{facultyUserId}/students	Authenticated	Assigned students.
Review Projects	GET	/deadline-rules/type/Project	Logged in	Deadline display.
Approve/Reject Project	PUT	/projects/mongo/review/{id}?status=&feedback=&grade=	FACULTY/ADMIN	Project review.
Download project	GET	/projects/download/{fileName}	Public	Open file.
Verify Certificates	GET	/deadline-rules/type/Certificate	Logged in	Deadline display.
Approve/Reject Certificate	PUT	/certificates/{id}/verify?status=&remarks=	FACULTY/ADMIN	Certificate verification.
Download certificate	GET	{certificate.fileURL}	Public	Open file.
Profile load	GET	/users/faculty/profile/{userId}	FACULTY/ADMIN	Faculty details.
Profile Save	PUT	/users/faculty/profile/{userId}	FACULTY/ADMIN	Update profile.
Notifications	GET	/users/notifications/{userId}	Logged in	Load notifications.
Mark read	PUT	/users/notifications/read/{id}	Logged in	Mark notification read.
Delete notification	DELETE	/users/notifications/{id}	Logged in	Delete notification.

Module 3 - Admin APIs

Button/screen action	Method	API	Role/Auth	Purpose
Dashboard	GET	/users/admin/analytics	ADMIN	Analytics.
Dashboard	GET	/users/audit-logs	ADMIN	Audit logs.
Manage Users	GET	/users	ADMIN	All users.
View student profile	GET	/users/student/profile/{id}	STUDENT/ADMIN	Student profile.
View faculty profile	GET	/users/faculty/profile/{id}	FACULTY/ADMIN	Faculty profile.
Edit user	PUT	/users/{id}	Logged in	Update user/profile.
Create user	POST	/auth/register	Public in config	Register user.
Post-create profile	PUT	/users/{createdUser.id}	Logged in	Extra profile fields.
Bulk Upload	POST	/users/bulk-upload	ADMIN	Excel import.
View Submissions	GET	/admin/projects	ADMIN	All projects.
View Submissions	GET	/admin/certificates	ADMIN	All certificates.
Settings load	GET	/deadline-rules	Logged in	All deadline rules.
Settings load	GET	/users/settings	Logged in	Admin settings.
Save Settings	POST	/users/settings	ADMIN	Save settings.
Save Deadline	POST	/deadline-rules	ADMIN	Create/update deadline.
Delete Deadline	DELETE	/deadline-rules/{id}	ADMIN	Delete deadline.
Faculty dropdown	GET	/users/filter/role?role=FACULTY	Logged in	Load faculty.
Assign Faculty	PUT	/users/assign-faculty/{id}?subject=	ADMIN	Assign subject.
Assign Student	PUT	/users/admin/assign-student?studentUserId=&facultyUserId=	ADMIN	Assign one student.
Unassign Student	PUT	/users/admin/unassign-student?studentUserId=	ADMIN	Remove assignment.
Assign by dept	PUT	/users/admin/assign-students-by-department?department=§ion=&facultyUserId=	ADMIN	Bulk assign.
Students by dept	GET	/users/students/departement/{department}	ADMIN	Department students.
Admin notifications	GET	/users/notifications/{userId}	Logged in	Load notifications.
Mark read	PUT	/users/notifications/read/{id}	Logged in	Mark notification read.
Delete notification	DELETE	/users/notifications/{id}	Logged in	Delete notification.
Admin dashboard ctrl	GET	/admin/dashboard	ADMIN	Admin counts.
Admin delete user	DELETE	/admin/users/{id}	ADMIN	Delete user.

Controller Base Paths

Controller	Base path	Purpose
UserController	/users	Login, forgot/reset password, users, profiles, settings, notifications, assignment.
AuthController	/auth	Registration.
ProjectController	/projects	Upload/download, project CRUD, review, filters, analytics.

Controller	Base path	Purpose
CertificateController	/certificates	Certificate CRUD and verification.
DeadlineRuleController	/deadline-rules	Deadline rule CRUD.
AdminController	/admin	Admin users/projects/certificates/dashboard.
LandingController	/landing	Button click tracking for Inspect/Network visibility.
DashboardController	/dashboard	Dashboard counts.